

Silhouette Coefficient: From First Principles

Why Accuracy Cannot Be Used

In supervised learning, we can compare predictions with true labels. That is why metrics like Accuracy, Precision, Recall, MAE, and RMSE make sense.

In clustering, we usually do not have true labels. K-Means may create clusters, but there is no answer key telling us which cluster each point should belong to.

Silhouette Coefficient helps evaluate clustering quality without needing true labels.

The Main Idea

A good cluster should satisfy two conditions. Points inside the same cluster should be close to each other. Points in different clusters should be far from each other.

Silhouette Coefficient measures both ideas at the same time.

It asks: is this point closer to its own cluster than to the nearest other cluster?

Intra-Cluster Distance

Intra-cluster distance measures how close a point is to other points inside its own cluster.

For a point i , this value is called $a(i)$.

If $a(i)$ is small, the point fits well inside its cluster.

If $a(i)$ is large, the point is far from its own cluster members.

Inter-Cluster Distance

Inter-cluster distance measures how close a point is to the nearest different cluster.

For a point i , this value is called $b(i)$.

If $b(i)$ is large, the point is far away from other clusters, which is good.

If $b(i)$ is small, the point is close to another cluster, which means the cluster assignment may be unclear.

Formula

The Silhouette Coefficient for one point is:

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

Here, $a(i)$ is the average distance from point i to points in its own cluster. The value $b(i)$ is the average distance from point i to points in the nearest other cluster.

The denominator keeps the score between -1 and 1.

Interpretation

A silhouette score close to 1 means the point is well matched to its own cluster and far from other clusters.

A score close to 0 means the point is near a boundary between clusters.

A score below 0 means the point may be assigned to the wrong cluster because it is closer to another cluster than to its own.

For the whole clustering result, we usually take the average silhouette score across all points.

Worked Example

Suppose a point has $a(i) = 2$ and $b(i) = 8$.

$$s(i) = \frac{8 - 2}{\max(2, 8)} = \frac{6}{8} = 0.75$$

This is a good score because the point is much closer to its own cluster than to another cluster.

Now suppose $a(i) = 6$ and $b(i) = 5$.

$$s(i) = \frac{5 - 6}{\max(6, 5)} = \frac{-1}{6} = -0.167$$

This is a bad sign because the point is closer to another cluster than to its assigned cluster.

Comparison with the Elbow Method

The Elbow Method uses inertia and helps choose K by looking for where improvement slows down. It is simple but sometimes unclear.

Silhouette Coefficient directly measures how well-separated the clusters are. It can compare different K values by checking which K gives a better average silhouette score.

Both methods are useful. The Elbow Method focuses on compactness. Silhouette focuses on compactness and separation.

When to Use It and Limitations

Use Silhouette Coefficient when you want to evaluate clustering without true labels. It is especially useful when comparing different values of K .

It can be expensive for very large datasets because it uses distances between points. It also works best when distance is a meaningful way to compare examples.

A high silhouette score is useful evidence, but it should not replace human judgment. Clusters should also make sense for the real problem.